

INTERNSHIP REPORT

Glencore Information Services Pvt Ltd

GLENCORE

Submitted by: Adithiyaa D
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ACKNOWLEDGMENT

I would like to sincerely thank Glencore for providing me with the opportunity to work as a C# and .NET Web Development Intern. Securing this internship through off-campus efforts has been a fulfilling and motivating experience, as it allowed me to independently step into a real-world software development environment beyond academics.

I am deeply grateful to my project mentor, Mr. Manoj and Mr. Narayanan, and the entire Glencore team for their constant guidance and support. Their approachability and collaborative mindset created an encouraging work environment that made the learning process both effective and enjoyable. The professional culture at Glencore significantly strengthened my confidence and technical competence in full-stack application development using the .NET ecosystem.

Through this internship, I gained hands-on experience with industry-relevant technologies such as C#, ASP.NET Core, Entity Framework, SQL databases, Angular and RESTful APIs. I also developed a strong understanding of key software engineering concepts including application architecture, API testing, database integration, and deployment workflows. Working on a project from the ground up enabled me to face practical challenges and develop efficient, scalable solutions.

I would also like to acknowledge the support and knowledge shared by the employees and fellow interns at Glencore, whose collaborative spirit greatly enriched my learning experience.

Lastly, I extend my heartfelt thanks to my family and close peers for their continuous encouragement throughout this journey. This internship has played a vital role in my growth as a developer and will strongly influence my future professional pursuits.

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1. INTRODUCTION

Academic projects play a vital role in bridging the gap between theoretical knowledge and real-world application. As part of my academic and professional learning journey, I designed and developed a web-based Customer Management System using AG Grid integrated with AI-powered data interaction features. This project focuses on building an intelligent, scalable and user-friendly data management interface suitable for enterprise environments.

The project leverages Angular, AG Grid, and RESTful APIs, combined with AI-driven filtering, sorting, and aggregation, to provide an advanced data visualization and management solution. Through this project, I gained hands-on experience in enterprise-grade frontend development, component-driven architecture, API integration, and intelligent user interaction design.

Purpose of the Internship

The primary purpose of this project was to design and implement a high-performance data grid system capable of managing large volumes of customer records efficiently while enhancing usability through AI-assisted interactions.

The project aims to:

- Provide an intuitive interface for viewing, filtering, sorting, and editing customer data.
- Integrate AI to allow users to interact with tabular data using natural language prompts.
- Demonstrate enterprise-level frontend practices using Angular and AG Grid.
- Improve data transparency, efficiency, and decision-making through real-time insights.
- Simulate real-world enterprise dashboards commonly used in business and analytics systems.

Overview of the Company

Glencore is a globally recognized diversified natural resource company, operating in over 35 countries. It holds a prominent position in mining, energy, metals, and agricultural commodities. While its core business revolves around commodity trading and industrial operations, Glencore also maintains a robust technological foundation to support its large-scale global activities.

The company fosters a professional yet flexible work culture that encourages collaboration, innovation, and continuous learning. The environment is welcoming and low-pressure, which helped me adapt quickly as an intern. Employees, from experienced developers to new joiners, were approachable and supportive, offering valuable insights into both software development and workplace ethics.

During my time at Glencore, I gained exposure to its technical ecosystem, which includes modern tools like Angular, C#, and Swagger, along with various cloud services. These technologies are used to build scalable, enterprise-grade platforms that support internal workflows and customer-facing applications. Although my primary work involved a MERN stack project, I got the chance to understand how their main tech stack functions in enterprise environments.

My mentor and teammates further enhanced my experience by introducing me to the fundamentals of .NET Core, Entity Framework, C#, and Angular. He also walked me through one of Glencore's internal projects named FinOps, built using vertical slicing and microservices architecture. The project demonstrated how scalable services are structured, and how Swagger plays a role in effective API testing and documentation.

Tools Used

During my internship at Glencore, I used a variety of development, database, and testing tools that contributed significantly to the successful completion of the Oil Booking System. These tools enhanced my understanding of modern software development workflows and improved my productivity:

Frontend

- **Angular (Standalone Components)** – For modular, scalable UI development.
- **AG Grid Community & Enterprise** – For advanced data grid functionalities.
- **TypeScript** – Strongly typed development for reliability and maintainability.
- **CSS & Custom Styling** – For responsive and professional UI design.

Backend & AI Integration

- **RESTful APIs** – For customer data retrieval and updates.
- **Gemini API** – Converts natural language queries into structured grid operations.
- **HTTP Client (Angular)** – For API communication.

Development & Testing Tools

- **Visual Studio Code** – Primary development environment.
- **SQL Server Management Studio** – Used to manage the customer database.
- **Visual Studio** – Primary development environment.
- **Swagger UI** – API testing and documentation.
- **Postman** – API validation and debugging.
- **Git & GitHub** – Version control and project management.
- **Browser DevTools** – Debugging rendering, network, and performance issues.

2. OBJECTIVE

The primary objective was to modernize how internal users interact with trading data. Instead of manually clicking through dozens of filters, the goal was to create an interface where non-technical users could simply ask the data questions.

Goals I Aimed to Achieve During the Internship

1. Understand Enterprise-Level Data Management

- Learn how large datasets are handled efficiently in enterprise applications.
- Implement scalable UI components capable of rendering thousands of records.

2. Master AG Grid for Advanced Data Visualization

- Utilize AG Grid features such as pagination, filtering, sorting, inline editing, and transactions.
- Apply column definitions, default column configurations, and grid APIs effectively.

3. Integrate AI with Frontend Applications

- Enable AI-driven filtering, sorting, and aggregation using natural language prompts.
- Translate AI responses into AG Grid-compatible filter and sort models.

4. Improve Frontend–Backend Integration

- Consume REST APIs to fetch and update customer data.
- Maintain synchronization between grid state and backend data.

5. Enhance Problem-Solving and Debugging Skills

- Resolve real-world issues related to grid theming, performance, and state handling.
- Debug Angular lifecycle, rendering, and styling conflicts.

3. COMPANY PROFILE

About Glencore

Glencore is one of the world's largest globally diversified natural resource companies, operating across over 35 countries. It is a leading integrated producer and marketer of commodities such as oil, metals, minerals, and energy products. The company plays a crucial role in the global supply chain by sourcing, producing, refining, transporting, and marketing essential raw materials that advance everyday life.

While Glencore is globally recognized for its role in the commodity and mining sectors, it also has strong internal IT and software teams that support its operational logistics, resource tracking, and enterprise-level system management.

Tech Stack and Domain

During my internship in Glencore's software and web development division, I had the opportunity to explore the company's technical environment, which includes a mix of enterprise-grade tools and modern web development technologies.

- Primary Technologies in Use:
 - Frontend: Angular(mainly used for internal web applications)
 - Backend: C#, .NET framework
 - API Testing & Documentation: Swagger
 - Database Systems: SQL Server
 - Other Tools & Practices: RESTful API development,microservices architecture, version control systems, and continuous integration tools.
- Domain:
 - The company primarily operates in the commodity trading and natural resource domain. The IT department supports systems for trading platforms, inventory management, logistics, compliance, and business intelligence.

Working Culture

Glencore fosters a professional and efficient work culture that encourages both individual responsibility and team collaboration. As an intern, I found the work environment to be:

- Supportive and learning-oriented – Team members were open to questions and often provided guidance and feedback.
- Process-driven – Projects followed structured workflows with clear development cycles, API standards, and review processes.
- Flexible and respectful – There was a healthy balance between tasks, learning time, and interaction with other employees, ensuring a stress-free yet productive experience.

4. PURPOSE OF THE PROJECT

1. Democratizing Data Access through Natural Language

Traditional data tools often require users to understand complex filtering logic. The purpose of this system is to allow non-technical users to query data using plain English. The project aims to remove the friction between thinking of a question and seeing the answer. Instead of manually clicking 15 different dropdown menus, the user's intent is instantly translated into executable code that configures the grid.

2. Enhancing Operational Efficiency and Speed

The project is designed to drastically reduce the time spent on report configuration. Automated state management means complex views that previously took minutes to set up manually are now generated instantly.

3. Intelligent & Dynamic Data Visualization

The system aims to intelligently organize data hierarchies. If a user asks about "monthly performance," the grid automatically groups rows by Month and aggregates values by Sum, eliminating the manual effort of setting up pivot tables.

4. Standardizing Data Interpretation

Consistent querying logic by using a centralized AI model to interpret queries, the project ensures that business logic is applied consistently across the organization. Manual data filtering is prone to human error. The AI system resets and applies exact specifications for every new query, ensuring the data displayed is exactly what was requested.

5. Modernizing the Enterprise User Experience (UX)

The project seeks to move away from static, rigid dashboards toward a conversational UI. This aligns the application with modern user expectations set by tools like Gemini, making internal enterprise software feel modern, intuitive, and engaging. Beyond simple filtering, the underlying purpose is to establish a technical foundation (Angular + AG Grid + Vector Context) that can support future advanced features, such as predictive analytics and automated anomaly detection in trading patterns.

5. PROJECT OVERVIEW

Project Title

AI-Powered Customer Management System – Web Application

Project Summary

The Oil Procurement System is a full-stack web application developed during a 1-month internship at Glencore to simplify the oil ordering process for both customers and administrators. It enables secure user registration and login, product browsing, order placement, and tracking. Admins can manage inventory, approve/reject orders, and view customer data. The system features real-time updates, role-based dashboards, and automated PDF invoice generation using modern MERN stack technologies.

Technology Stack

Frontend

- **Angular** – For modular, scalable UI development.
- **AG Grid Community & Enterprise** – For advanced data grid functionalities.
- **TypeScript** – Strongly typed development for reliability and maintainability.
- **CSS & Custom Styling** – For responsive and professional UI design.

Backend & AI Integration

- **RESTful APIs** – For customer data retrieval and updates.
- **Gemini API** – Converts natural language queries into structured grid operations.
- **HTTP Client (Angular)** – For API communication.
- **Framework:** .NET
- **Language:** C#

Database

- **SQL** using **SQL Server Management Studio**

API Testing

- **POSTMAN** for API testing
- **Swagger UI** for API testing

Implemented Functionality

- **Filter:** Text-to-Filter conversion (NLP to JSON).
- **Sorting:** Text-to-Sort conversion (NLP to JSON).
- **Column Visibility:** Text-to-Visibility conversion (NLP to JSON).
- **Column Resizing:** Text-to-Resizing conversion (NLP to JSON).
- **Data Aggregation:** Statistics based on numeric fields.
- **Data Export:** Exporting specific views to CSV/Excel based on AI-defined scopes.
- **Chat Interface:** A side-panel chatbot that is context-aware of the data currently visible in the grid.

Challenges Faced & Solutions

1. Frontend–Backend Sync

- Issue: Edited data not reflecting immediately.
- Fix: Used AG Grid transactions and API-based updates for seamless synchronization.

2. AG Grid Theming Conflicts

- Issue: Conflicts between legacy CSS themes and the new theming API.
- Fix: Adopted legacy themes consistently and removed incompatible configurations.

3. AI Hallucinations

- Issue: The AI would sometimes invent column names that didn't exist in the database.
- Fix: I implemented a "Schema Injection" technique where the prompt explicitly lists the allowed colIds, strictly constraining the AI's output.

4. AI Response Mapping

- Issue: Translating AI output into grid-compatible models.
- Fix: Designed a structured response format mapping AI output directly to AG Grid filter and sort models.

5. Latency

- Issue: AI responses took 2-3 seconds.
- Fix: Implemented optimistic UI updates and loading skeletons to keep the interface feeling responsive.

6. Large Dataset Rendering

- Issue: Performance concerns with large datasets.
- Fix: Implemented pagination and optimized grid rendering using AG Grid APIs.

6. OUTPUT SCREENSHOTS

1. Dashboard



Customer Management

View, filter, sort and edit customer records with AI assistance

Q

Ask AI to filter, sort, or analyze data... (e.g., 'show inactive customers', 'sort by orders descending')

★ AI

I, ≡	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	Anil Joshi	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
2	Sanjay Singh	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	☒
3	Abhishek Chatterjee	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	☒
4	Rohit Ghosh	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	☒
5	Naveen Yadav	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	☒
6	Mahesh Kumar	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	☒
7	Meera Ghosh	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
8	Amit Bansal	amit.bansal8@gmail.com	9464293796	11/11/1967	37	☒
9	Ashwin Nair	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	☒
10	Meera Joshi	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒

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2. Filtering - Single and Multi



Customer Management
View, filter, sort and edit customer records with AI assistance

Q filter emails containing joshi

★ AI

	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	Anil Joshi	aniljoshi1@gmail.com	9995267406	03/12/1983	38	☒
10	Meera Joshi	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒
22	Harish Joshi	harish.joshi22@gmail.com	9186658063	31/08/2006	18	☐
26	Akash Joshi	akash.joshi26@gmail.com	9825010840	02/10/1990	4	☒
53	Divya Joshi	divya.joshi53@gmail.com	9854673810	25/09/2005	7	☒
76	Amit Joshi	amit.joshi76@gmail.com	9225819932	15/12/1993	4	☒
100	Rohit Joshi	rohit.joshi100@gmail.com	9607764374	17/02/2008	8	☒
101	Suresh Joshi	suresh.joshi101@gmail.c...	9245519651	15/12/1990	38	☒
138	Akash Joshi	akash.joshi138@gmail.com	9415676593	02/07/2000	15	☒
161	Sneha Joshi	sneha.joshi161@gmail.com	9268352722	23/04/1986	43	☒

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Customer Management
View, filter, sort and edit customer records with AI assistance

Q filter customer names which contains Abhishek and customer phone no ends with 8

★ AI

	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
676	Abhishek Aggarwal	abhishek.aggarwal676@...	9677519368	22/01/1986	41	☐
707	Abhishek Chatterjee	abhishek.chatterjee707@...	9956205018	24/02/2004	18	☒
805	Abhishek Chatterjee	abhishek.chatterjee805@...	9410891728	18/08/2000	30	☒

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3. Sorting - Single and Multi



Customer Management
View, filter, sort and edit customer records with AI assistance

Q

sort in descending order of dob

★ AI

	Customer Name	Customer Email	Customer Phone	Dob ↓	Total Orders	Is Active
110	Nisha Yadav	nisha.yadav110@gmail.c...	9225459323	26/12/2008	28	☒
436	Amit Nair	amit.nair436@gmail.com	9743648777	24/12/2008	13	☒
637	Varun Kumar	varun.kumar637@gmail.c...	9616550601	18/12/2008	13	☐
489	Suresh Iyer	suresh.iyer489@gmail.com	9317512599	10/11/2008	27	☒
901	Ravi Bansal	ravi.bansal901@gmail.com	9803686888	22/10/2008	11	☐
358	Rohit Joshi	rohit.joshi358@gmail.com	9115292610	22/10/2008	31	☐
457	Kavya Mehta	kavya.mehta457@gmail.c...	9765577253	12/09/2008	33	☒
540	Mahesh Kumar	mahesh.kumar540@gmai...	9227491725	31/08/2008	38	☒
898	Neha Naidu	neha.naidu898@gmail.com	9415170413	27/08/2008	10	☒
288	Arjun Bansal	arjun.bansal288@gmail.c...	9210712030	21/08/2008	40	☒

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Customer Management
View, filter, sort and edit customer records with AI assistance

Q

sort the customers in ascending order of phone no followed by ascending order of total orders

★ AI

	Customer Name	Customer Email	Customer ... 1 ↑	Dob	Total Orders 2 ↑	Is Active
145	Priya Naidu	priya.naidu145@gmail.com	9101867269	06/08/1997	37	☒
500	Suresh Joshi	suresh.joshi500@gmail.c...	9104550866	03/10/1967	24	☐
133	Rahul Banerjee	rahul.banerjee133@gmail...	9104741087	25/12/2003	11	☒
562	Priya Gupta	priya.gupta562@gmail.co...	9104832920	02/10/1972	43	☒
183	Rahul Aggarwal	rahul.aggarwal183@gmai...	9105078184	03/07/1970	30	☒
736	Amit Das	amit.das736@gmail.com	9105140199	20/03/1967	0	☒
122	Akash Kumar	akash.kumar122@gmail.c...	9107399129	25/09/2003	10	☒
420	Divya Iyer	divya.iyer420@gmail.com	9107863570	20/02/2007	14	☒
977	Nisha Yadav	nisha.yadav977@gmail.c...	9109284573	21/05/1975	42	☐
717	Deepak Malhotra	deepak.malhotra717@g...	9109706832	20/09/1973	48	☐

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4. Column Visibility

- Hide certain cols



Customer Management

View, filter, sort and edit customer records with AI assistance

hide column customer name

★ AI

ID	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
2	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	☒
3	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	☒
4	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	☒
5	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	☒
6	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	☒
7	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
8	amit.bansal8@gmail.com	9464293796	11/11/1967	37	☒
9	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	☒
10	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒

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- Show certain cols



Customer Management

View, filter, sort and edit customer records with AI assistance

show column customer name

★ AI

Customer Name	ID	Customer Email	Customer Phone	Dob	Total Orders	Is Active
Anil Joshi	1	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
Sanjay Singh	2	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	☒
Abhishek Chatterjee	3	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	☒
Rohit Ghosh	4	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	☒
Naveen Yadav	5	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	☒
Mahesh Kumar	6	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	☒
Meera Ghosh	7	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
Amit Bansal	8	amit.bansal8@gmail.com	9464293796	11/11/1967	37	☒
Ashwin Nair	9	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	☒
Meera Joshi	10	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒

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● Hide all cols

 **Customer Management**
View, filter, sort and edit customer records with AI assistance

Q hide all the columns

★ AI

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● Show all cols

 **Customer Management**
View, filter, sort and edit customer records with AI assistance

Q show all columns

★ AI

ID	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	Anil Joshi	aniljoshi1@gmail.com	9995267406	03/12/1983	38	✓
2	Sanjay Singh	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	✓
3	Abhishek Chatterjee	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	✓
4	Rohit Ghosh	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	✓
5	Naveen Yadav	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	✓
6	Mahesh Kumar	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	✓
7	Meera Ghosh	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	✓
8	Amit Bansal	amit.bansal8@gmail.com	9464293796	11/11/1967	37	✓
9	Ashwin Nair	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	✓
10	Meera Joshi	meera.joshi10@gmail.com	9744525888	26/08/1973	35	✓

5. Column Resizing

**Customer Management**
View, filter, sort and edit customer records with AI assistance

Q

set column size for id as 200 px

★ AI

ID	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	Anil Joshi	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
2	Sanjay Singh	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	☒
3	Abhishek Chatterjee	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	☒
4	Rohit Ghosh	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	☒
5	Naveen Yadav	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	☒
6	Mahesh Kumar	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	☒
7	Meera Ghosh	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
8	Amit Bansal	amit.bansal8@gmail.com	9464293796	11/11/1967	37	☒
9	Ashwin Nair	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	☒
10	Meera Joshi	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒

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6. Aggregation

**Customer Management**
View, filter, sort and edit customer records with AI assistance

Q

sum and avg of total orders

★ AI

ID	Customer Name	Customer Email	Customer Phone	Dob	Total Orders	Is Active
1	Anil Joshi	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
2	Sanjay Singh	sanjay.singh2@gmail.com	9343574918	20/07/2003	28	☒
3	Abhishek Chatterjee	abhishek.chatterjee3@g...	9290635531	21/08/1979	41	☒
4	Rohit Ghosh	rohit.ghosh4@gmail.com	9777474883	28/10/1984	15	☒
5	Naveen Yadav	naveen.yadav5@gmail.com	9300463722	24/08/1999	39	☒
6	Mahesh Kumar	mahesh.kumar6@gmail.c...	9787268693	13/10/2003	28	☒
7	Meera Ghosh	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
8	Amit Bansal	amit.bansal8@gmail.com	9464293796	11/11/1967	37	☒
9	Ashwin Nair	ashwin.nair9@gmail.com	9188971820	29/03/1973	23	☒
10	Meera Joshi	meera.joshi10@gmail.com	9744525888	26/08/1973	35	☒

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Data Insights

TOTALORDERS_SUM

24887

TOTALORDERS_AVG

24.887



Customer Management

View, filter, sort and edit customer records with AI assistance

Q sum and avg of customers with name containing Anil

★ AI

I. ⌵	Customer Name ⌵	Customer Email ⌵	Customer Phone ⌵	Dob ⌵	Total Orders ⌵	Is Active ⌵
1	Anil Joshi	anil.joshi1@gmail.com	9995267406	03/12/1983	38	☒
13	Anil Malhotra	anil.malhotra13@gmail.c...	9367595492	07/11/2007	34	☒
77	Anil Verma	anil.verma77@gmail.com	9173248538	04/11/1970	24	☒
79	Anil Sharma	anil.sharma79@gmail.com	9280676505	27/10/1967	25	☒
104	Anil Yadav	anil.yadav104@gmail.com	9459547007	21/12/2000	45	☒
105	Anil Singh	anil.singh105@gmail.com	9535970456	18/02/1998	20	☒
139	Anil Nair	anil.nair139@gmail.com	9952070413	04/09/1983	1	☒
147	Anil Verma	anil.verma147@gmail.com	9767395651	30/01/1987	27	☒
178	Anil Kumar	anil.kumar178@gmail.com	9603532607	23/09/1984	48	☒
186	Anil Joshi	anil.joshi186@gmail.com	9703618206	24/06/1981	22	☒

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Data Insights

TOTALORDERS_SUM
1227

TOTALORDERS_AVG
29.214285714285715



Customer Management

View, filter, sort and edit customer records with AI assistance

Q sum, count, min, max and avg of customers with dob greater than 03/12/2005

★ AI

I. ⌵	Customer Name ⌵	Customer Email ⌵	Customer Phone ⌵	Dob ⌵	Total Orders ⌵	Is Active ⌵
7	Meera Ghosh	meera.ghosh7@gmail.com	9180493726	21/07/2008	26	☒
13	Anil Malhotra	anil.malhotra13@gmail.c...	9367595492	07/11/2007	34	☒
22	Harish Joshi	harish.joshi22@gmail.com	9186658063	31/08/2006	18	☐
32	Divya Naidu	divya.naidu32@gmail.com	9531763302	01/04/2006	32	☒
40	Neha Chatterjee	neha.chatterjee40@gmail...	9391662441	02/02/2008	44	☐
43	Sneha Bansal	sneha.bansal43@gmail.c...	9127671387	14/07/2007	30	☒
53	Divya Joshi	divya.joshi53@gmail.com	9854673810	25/09/2005	7	☒
58	Meera Reddy	meera.reddy58@gmail.co...	9371750084	18/11/2007	29	☒
100	Rohit Joshi	rohit.joshi100@gmail.com	9607764374	17/02/2008	8	☒
109	Amit Singh	amit.singh109@gmail.com	9502752046	31/10/2006	20	☒

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Data Insights

TOTALORDERS_SUM
2451

TOTALORDERS_COUNT
94

TOTALORDERS_MIN
0

TOTALORDERS_MAX
49

TOTALORDERS_AVG
26.074468085106382

7. INTERNSHIP EXPERIENCE

My 4-week internship at Glencore as a Web Development Intern offered a valuable opportunity to dive into the intersection of modern web development and Artificial Intelligence.

During the first week, I focused on understanding the company's data needs. I learned that in trading, speed and data clarity are paramount. My mentor introduced me to AG Grid, explaining why standard HTML tables are insufficient for enterprise data. I spent time reading documentation and building small prototypes to understand features like "Row Virtualization" and "Server-Side Row Models."

From the second week onwards, I began working on the AI-Powered Dashboard. This was a challenging shift from standard CRUD applications. I had to learn how to prompt-engineer an AI model to output valid JSON code that could control the grid. I designed a React architecture where the user's chat input was sent to a backend Node.js server; the server processed the request with the LLM and returned a "Grid State Object."

I implemented features where the AI could control visibility, sorting, and filtering. For example, if a user typed "Reset view", the AI would return a command to clear all filters. Implementing this required a deep understanding of React's Context API to pass state between the Chat Component and the Grid Component.

This internship strengthened my understanding of Prompt Engineering as a development tool, not just a chatbot feature. It taught me how to constrain AI outputs to make them reliable for software control. The supportive environment at Glencore allowed me to experiment with these cutting-edge technologies, providing me with a unique portfolio project that stands out from standard web apps.

8. CONCLUSION

Interning at Glencore has been a defining moment in my early career. It shifted my perspective from simply "building websites" to "building intelligent tools." Working with AG Grid showed me the complexity of enterprise software, while integrating AI taught me about the future of user interfaces—where we move from clicking buttons to declaring intent.

The work culture at Glencore emphasized innovation. Even as an intern, I was encouraged to explore how AI could optimize their traditional workflows. This autonomy, combined with the technical rigor of working with financial/trading data, has prepared me significantly for a future in Full Stack Development and AI Engineering.

I would like to extend my heartfelt gratitude to Glencore for providing this platform. The experience of merging robust industrial tools with modern AI capabilities was intellectually stimulating and professionally rewarding.

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6. <https://chatgpt.com/>
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